

whereas,

$$\begin{aligned}
 e^{hA} e^{hB} &= (\text{Id} + hA + \frac{h^2 A^2}{2} + \frac{h^3 A^3}{6} + \mathcal{O}(h^4)) \\
 &\quad \times (\text{Id} + hB + \frac{h^2 B^2}{2} + \frac{h^3 B^3}{6} + \mathcal{O}(h^4)) \\
 &= \text{Id} + h(A + B) + \frac{h^2}{2}(2AB + A^2 + B^2) \\
 &\quad + \frac{h^3}{6}(A^3 + B^3 + 3AB^2 + 3A^2B) + \mathcal{O}(h^4).
 \end{aligned}$$

So the difference is

$$\begin{aligned}
 e^{hA} e^{hB} - e^{h(A+B)} &= \frac{h^2}{2}(AB - BA) \\
 &\quad + \frac{h^3}{6}(2AB^2 + 2A^2B - BA^2 - BAB - B^2A - ABA) + \mathcal{O}(h^4) \\
 &= \frac{h^2}{2}[A, B] + \mathcal{O}(h^3),
 \end{aligned}$$

where  $[A, B] = AB - BA$  is the commutator of  $A$  and  $B$ .

For a splitting method, we have, replacing  $A$  by  $\mathcal{L}_{H_1}$  and  $B$  by  $\mathcal{L}_{H_2}$ ,

$$e^{h\mathcal{L}_{H_1}} e^{h\mathcal{L}_{H_2}} - e^{h\mathcal{L}_H} = \frac{h^2}{2}[\mathcal{L}_{H_1}, \mathcal{L}_{H_2}] + \mathcal{O}(h^3).$$

Observe that, for any real-valued functions  $f$  and  $H$  of phase space,

$$\mathcal{L}_H f = \nabla f^T \mathbf{J} \nabla H = \{f, H\}.$$

Hence

$$\mathcal{L}_{H_1} \mathcal{L}_{H_2} f = \{\{f, H_2\}, H_1\},$$

and, using this and skew-symmetry of the Poisson bracket,

$$[\mathcal{L}_{H_1}, \mathcal{L}_{H_2}]f = \{\{f, H_2\}, H_1\} - \{\{f, H_1\}, H_2\} = \{\{f, H_2\}, H_1\} + \{\{H_1, f\}, H_2\}.$$

By the Jacobi identity,

$$\{\{f, H_2\}, H_1\} + \{\{H_2, H_1\}, f\} + \{\{H_1, f\}, H_2\} = 0.$$